Response to Reviewers

Manuscript title (revised):
Benchmarking deep and hybrid quantum-classical models for Gallo-Roman ceramic sherd classification: a reproducible evaluation framework

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We would like to thank the Editor and the reviewers for their careful reading of our manuscript and for their constructive and detailed comments. We have substantially revised the manuscript in response. Compared with the originally reviewed version, the revised manuscript now reframes the contribution explicitly as a methodological and reproducible evaluation framework rather than as an architectural novelty claim, clarifies the analytical distinction between crop-based classification and end-to-end detection, redefines the imbalanced and balanced scenarios more rigorously, corrects and recomputes the end-to-end RT-DETR analysis using a unified evaluation protocol, adds threshold-sensitivity, per-class AP50, and multi-seed robustness analyses, expands the archaeological and heritage-science interpretation of the results, adds learning curves and an evaluation-metrics glossary in the Supplementary Materials, and makes the code publicly available.

The manuscript title has also been revised to reflect the actual scope of the work:
Benchmarking deep and hybrid quantum-classical models for Gallo-Roman ceramic sherd classification: a reproducible evaluation framework.

Below we provide a point-by-point response to each comment. For clarity, we indicate both the substance of each revision and the precise location in the revised manuscript where the corresponding changes were made.

Reviewer 1

Comment 1

Reviewer comment

The current disciplinary contribution is not sufficiently clear, and the methodological novelty cannot be verified. The core contribution at present reads as a benchmarking-based summary of experience, lacking a clearly defined production of new knowledge. The authors should explain what innovation this brings to heritage science, not only from a heritage perspective but also in terms of methodological increment.

Authors' response

We thank the reviewer for this central comment. We agree that the originally reviewed version did not make the disciplinary and methodological contribution sufficiently explicit. We have therefore reframed the manuscript so that its primary contribution is now stated clearly as methodological rather than architectural.

In the revised manuscript, we now emphasise that the study does not introduce a new model architecture. Instead, it establishes a controlled and reproducible evaluation framework for comparing two computer-vision paradigms derived from the same annotated corpus: conditioned classification on ground-truth crops and end-to-end detection on full images. We also clarify that the transferable output of the study is the protocol itself: matched image-level splitting, task-specific but harmonised evaluation, explicit handling of class imbalance, and explicit analysis of the detector operating point.

To make this repositioning visible from the outset, we revised not only the Introduction but also the title and abstract. The revised Discussion now also states explicitly that the same archaeological problem leads to different conclusions depending on task formulation, and that this methodological shift is itself a substantive contribution for heritage-science evaluation.

Changes made in the manuscript

- Title revised to: "Benchmarking deep and hybrid quantum-classical models for Gallo-Roman ceramic sherd classification: a reproducible evaluation framework."
- Abstract rewritten to foreground the evaluation-framework contribution and the importance of task formulation.
- Introduction revised, especially the paragraphs beginning "This study proposes a reproducible evaluation framework..." and "This study does not propose a new model architecture...".
- Methods revised in the subsection "Methodological pipeline and controlled evaluation protocol".
- Discussion revised, especially the opening methodological paragraphs beginning "Beyond model ranking alone..." and "A key methodological outcome of this work...".

Comment 2

Reviewer comment

The comparison between GT-crop classification and end-to-end detection is valuable, but it remains a parallel validation of established paradigms. The authors do not provide a unique argument for a new evaluation exemplar or a generalisable protocol.

Authors' response

We thank the reviewer for this important remark. In the revised manuscript, we now justify this comparison more explicitly as a comparison between two analytically distinct situations rather than as a parallel benchmark of established architectures.

We explain that crop-based classification corresponds to a conditioned recognition task in which the sherd has already been isolated and tightly framed, which matches a diagnostic stage of ceramological work. By contrast, end-to-end detection corresponds to a full decision pipeline in which localisation and classification must be solved jointly on full images containing multiple fragments, which matches the initial sorting stage of archaeological practice.

We also now state more clearly that the value of the comparison lies in the protocol: both paradigms are derived from the same source corpus, share the same image-level split, and are analysed under matched reporting conditions. This is the basis on which we argue that the study proposes a reusable evaluation design rather than a simple side-by-side comparison.

Changes made in the manuscript

- Introduction revised, especially the paragraph beginning “To analyse model behaviour under realistic archaeological conditions, we compare two complementary computer vision paradigms...”.
- Introduction revised to link each paradigm explicitly to a distinct stage of the ceramologist’s workflow.
- Methods revised in the subsection “Methodological pipeline and controlled evaluation protocol” to clarify the shared image-level split and the controlled comparison logic.

Comment 3

Reviewer comment

The inclusion of QCNN/QCNN-VQE currently appears to treat the quantum head as a candidate classification head for comparison, but it does not yield conclusions that constitute new insights. The authors need to explain under what data conditions, noise conditions, or other constraints the quantum head offers stable advantages.

Authors’ response

We thank the reviewer for this important comment. We have moderated and clarified the role of the quantum component throughout the manuscript.

In the revised version, QCNN and QCNN-VQE are no longer presented as evidence of a general quantum advantage. They are now framed as exploratory hybrid CNN–PQC classifiers included in the benchmark to assess whether a highly compact quantum-classical decision head can remain competitive under controlled archaeological classification conditions. We explicitly state that all quantum experiments were conducted in classical simulation, and that the present study does not support any claim of quantum computational advantage.

We also now state more clearly that no stable superiority over compact classical baselines is claimed. Their interest in this benchmark is instead linked to compactness and efficiency trade-offs. Questions related to realistic hardware noise, scaling, or possible data-regime-specific advantages are now clearly identified as outside the scope of the present manuscript and left for future work.

Changes made in the manuscript

- Introduction revised, especially the paragraph beginning “The models evaluated include reference convolutional classifiers...”.
- Methods revised in the subsection “Architecture of QCNN and QCNN-VQE models”.
- Discussion revised in the paragraphs beginning “With regard to PQC models...” and “QCNN-based models emerge here...”.

Comment 4

Reviewer comment

The authors should more clearly explain the paper’s knowledge increment for the heritage science problem itself. At present, the work does not produce new academic conclusions or findings regarding the scientific meaning of the sigillata/OW/RW technological classifications, their material differences, or the sources of texture; instead, it largely treats the task as a three-class classification problem.

Authors’ response

We thank the reviewer for this important comment. We have expanded the archaeological and heritage-science interpretation so that the manuscript no longer treats the task as a purely abstract three-class benchmark.

First, we now make explicit that terra sigillata is consistently the most recognisable category across model families and evaluation settings, and we relate this result to its relative technological and visual homogeneity. Second, we discuss more explicitly that most residual errors are concentrated in OW/RW confusions, and we link this difficulty to the visual and technological proximity of these two categories, including their broader workshop diversity, firing variability, and taphonomic alteration.

We also strengthened the practical heritage-science interpretation of crop-based classification. In the revised Discussion, crop-based recognition is no longer presented as merely a simplified computer-vision task; it is interpreted as a realistic analytical situation that corresponds to the individual examination of already isolated fragments in ceramological practice. More broadly, the revised manuscript now argues that RGB-image AI does not capture all ceramic distinctions equally well, and that this asymmetry is itself informative for heritage workflows.

Changes made in the manuscript

- Results expanded in the subsection “Complementary analysis and typical errors”.

- Discussion expanded in the paragraphs beginning “From an archaeological perspective...”, “From a heritage science perspective...”, and “A second observation concerns the practical relevance of crop-based classification...”.
- Discussion also expanded in the paragraphs beginning “From a heritage-science perspective, the most consistent result of this benchmark...” and “Beyond model comparison...”.

Comment 5

Reviewer comment

This is a major issue. The reported mAP of RT-DETR is not low, yet under a fixed threshold the end-to-end F1 for OW/RW is close to 0. This combination is highly unusual, raising concerns about threshold selection, class mapping, and post-processing consistency. The authors must provide an explanation; otherwise, the architectural conclusions become untenable. It is recommended to report per-class AP for RT-DETR, and provide PR curves or a confidence-threshold sweep curve, marking the chosen point at conf = 0.25.

Authors’ response

We thank the reviewer for identifying this critical issue. After re-examining the original end-to-end evaluation pipeline, we found inconsistencies in the initial alignment between detector outputs and ground-truth image identifiers. We therefore recomputed the detector-specific end-to-end analyses using a unified one-to-one IoU-based matching protocol and consistent post-processing settings across models.

As a result, the revised manuscript no longer reports near-zero OW/RW F1 scores for RT-DETR. Instead, RT-DETR now remains clearly weaker than the YOLO-based detectors at the fixed benchmark operating point, but with interpretable and internally consistent results. We also now distinguish explicitly between two metric families: COCO-style ranking metrics (AP, mAP), which integrate performance across thresholds, and fixed-threshold decision metrics (end-to-end macro-F1), which are more sensitive to over-prediction, calibration, and inter-class confusion at a chosen operating point.

In direct response to the reviewer’s recommendation, we added a confidence-threshold sweep with the benchmark point marked at conf = 0.25, a fixed-threshold diagnostic analysis for RT-DETR, and per-class AP50 values under the COCO protocol. These additions show that RT-DETR retains acceptable localisation-oriented AP values, but that this does not translate into equally strong class-balanced decisions at a fixed threshold because of higher false-positive rates and persistent OW/RW confusion.

We also audited the class-index mapping across annotation conversion, detector training and evaluation (0 = terra sigillata, 1 = OW, 2 = RW), and added this clarification to the Evaluation metrics subsection of the revised manuscript.

Changes made in the manuscript

- Methods revised in the subsection “Evaluation metrics”, especially the paragraphs beginning “All end-to-end detection macro-F1 scores...” and “For end-to-end detection, we report standard detection metrics...”.
- Main benchmark detector tables updated (Tables 4 and 5) after recomputation with the unified end-to-end protocol.
- Results expanded with the new subsection “Confidence-threshold sensitivity and per-class detection behaviour”, including Figure 12 and the fixed-threshold diagnostic Table 6.
- Results expanded with the new subsection “Per-class detection performance and threshold sensitivity of RT-DETR”, including Table 8 (per-class AP50 values).
- Discussion revised to explain explicitly the divergence between ranking-based mAP and fixed-threshold macro-F1.
- Methods revised to state explicitly that class-index mapping was kept identical across annotation conversion, detector training and evaluation.

Comment 6

Reviewer comment

The current conclusions are mainly based on a single image-level 80/20 split and a single run. The lack of uncertainty estimation weakens statistical robustness. It is recommended to repeat experiments with multiple random seeds for several key models and report means and confidence intervals.

Authors’ response

We thank the reviewer for this important suggestion. To strengthen statistical robustness, we added a multi-seed analysis for key classification and detection models using seeds 42, 123, and 777. In the revised manuscript, performance is now reported as mean ± standard deviation together with 95% confidence intervals.

For classification models, we report macro-F1 across seeds. For detection models, we now report both standard COCO metrics (mAP50 and mAP50–95) and end-to-end macro-F1 at the fixed benchmark operating point, in full consistency with the revised evaluation protocol. These additional experiments confirm that the relative ranking of the architectures remains stable across seeds, while also showing that RT-DETR is more variable than the YOLO-based detectors.

Changes made in the manuscript

- Results expanded with the new subsection “Robustness across random seeds”.
- Table 9 added to report the multi-seed results for classification and detection, including mean ± SD and 95% confidence intervals.

- Discussion revised to state explicitly that the main benchmark conclusions remain stable across random initialisations.

Comment 7

Reviewer comment

The authors need to explain how generalisation is demonstrated. Data from a single location and a single acquisition workflow cannot support claims of a generalisable methodology.

Authors' response

We agree with the reviewer. We have therefore moderated the scope of our claims.

In the revised manuscript, we no longer present the reported numerical performance as evidence of empirical generalisation across archaeological sites, collections, or acquisition workflows. Instead, we state explicitly that the numerical results remain specific to the Amiens corpus and its acquisition conditions. The transferable contribution is now framed as the evaluation protocol itself: a controlled and reproducible design that can be reused on other archaeological datasets to compare conditioned classification and end-to-end detection under matched conditions.

Changes made in the manuscript

- Introduction revised, especially the paragraph beginning "This study does not propose a new model architecture..." and its final sentence on the corpus-specific scope of the numerical results.
- Discussion revised in the limitations paragraph beginning "Several limitations and extensions follow from this study...".

Comment 8

Reviewer comment

In the paper, balanced is mainly achieved via training-set resampling, but its meaning and implementation are not strictly equivalent across tasks. Moreover, in crop classification, the two versions do not have exactly the same validation-set size. Please provide a strict definition of "balanced": at what level is balancing performed, crops or full images? Does it introduce duplicated backgrounds or highly similar images?

Authors' response

We thank the reviewer for this important comment. We have rewritten the description of the imbalanced and balanced scenarios so that the meaning of "balanced" is now defined strictly and task by task.

In the revised manuscript, "balanced" refers only to the distribution of training samples used during optimisation; it does not describe a modification of the underlying archaeological corpus. For crop-based classification, balancing is defined at the crop level. For end-to-end detection, balancing is defined at the training-image level, by increasing the sampling and augmentation of images containing under-represented classes while preserving the original image structure. Validation is never augmented.

To answer the reviewer's question directly: train-only rebalancing may generate augmented variants of existing training crops or training images, which can therefore retain similar visual backgrounds or acquisition conditions, but these variants remain confined strictly to the training subset. Because splitting is performed at the source-image level before crop derivation or augmentation, no augmented duplicate or near-duplicate derived from a training image can enter validation.

We also now explain explicitly why the validation crop totals differ slightly between the two crop-based datasets (694 vs 693): the two crop datasets were derived separately after image-level splitting and filtering, but the image-level separation between training and validation is unchanged, so there is no leakage. To remove any remaining ambiguity, we harmonised the dataset tables and their captions so that full-image detection and crop-based classification are distinguished explicitly.

Changes made in the manuscript

- Methods fully revised in the subsection "Imbalanced and balanced scenarios", especially the paragraphs beginning "In this study, the term balanced refers only...", "For crop-based classification, balancing was performed...", and "For end-to-end detection, the validation set is identical...".
- Table 1 retained to summarise crop totals across scenarios and splits.
- Table 2 caption revised to describe explicitly the full-image detection datasets used in the imbalanced and balanced scenarios.
- Table 3 caption revised to describe explicitly the derived crop-classification datasets used in the imbalanced and balanced scenarios, including the explanation of the 694 vs 693 validation-crop difference.
- The corresponding Methods sentence was revised to read, in substance: the class distributions are reported in Table 2 for end-to-end detection and in Table 3 for crop-based classification.

Comment 9

Reviewer comment

The current citations and prior literature discussion are mostly used to show that methods exist, but the paper lacks discussion of debates around combining deep learning with heritage ceramics and sherd analysis. Consider adding key works on heritage ceramic recognition and intelligent archaeological imaging, and clearly state how this paper differs in task design, data, and evaluation. For imbalanced learning, more authoritative methodological papers should be cited rather than relying mainly on experiential explanation.

Authors' response

We thank the reviewer for this important recommendation. We have expanded and restructured the literature review so that the manuscript is now positioned more clearly within both the archaeological-ceramic literature and the methodological computer-vision literature.

More specifically, we added and integrated prior work on archaeological ceramic recognition and related heritage-imaging applications, and we clarified how the present study differs from prior work: it derives both tasks from the same source corpus, compares conditioned crop classification and end-to-end detection under matched image-level splits, and analyses both class imbalance and detector operating-point sensitivity. We also strengthened the methodological background on imbalanced learning by citing more authoritative references on imbalance-aware learning and by explaining more clearly why we chose a conservative train-only rebalancing strategy in this context.

Changes made in the manuscript

- Introduction expanded, especially the paragraphs beginning "Despite recent progress in computer vision applications...", "One widely used approach relies on one-stage detectors...", "DETR-type detectors approach object detection...", and "Another emerging direction concerns hybrid quantum–classical learning approaches...".
- Methods strengthened in the subsection "Managing class imbalance".
- Discussion expanded near the end to position the work relative to recent heritage-imaging studies, including pottery-documentation digitisation.

Reviewer 2

Comment 1

Reviewer comment

The only thing I feel is missing is that many highly specialised terms such as macro/class F1, mAP, AP@0.5/AP@[0.5:.95], COCO protocol etc, were used without definition in this paper. These terms have been commonly used in the field of computer vision and machine learning. However, considering the main audience in this journal are not specialist in computer vision and/or machine learning, missing these definitions would make the paper difficult to read. An appendix of the definition of these terms would make this paper much easier to understand for broader audience or at least proper references should be cited.

Authors' response

We thank the reviewer for this very helpful suggestion. We have revised the manuscript to improve accessibility for readers who are not specialists in computer vision.

In the revised manuscript, the "Evaluation metrics" subsection now defines the logic of the main metrics used in the paper. In addition, the Supplementary Materials now include a dedicated

"Evaluation Metrics" glossary that defines precision, recall, F1-score, class-wise F1, macro-F1, IoU, confidence threshold, one-to-one matching, AP, AP@0.5, AP@[0.5:0.95], mAP, the COCO evaluation protocol, and the distinction between mAP and fixed-threshold macro-F1.

Changes made in the manuscript

- Methods revised in the subsection "Evaluation metrics".
- Supplementary Materials expanded with a dedicated "Evaluation Metrics" section and glossary.

Comment 2

Reviewer comment

The validation procedure in this paper is appropriate but it may be further improved if the authors could consider testing multiple combinations of training and test sets and assess the reproducibility of the performances of the model. Such test would provide valuable insight into the generalization performance of the models. It would be also interesting to see how the sample size, i.e. different split ratio of training and test set, would impact the performance of the model, this will give important information about the generalization capability of the models. In summary, it would be highly valuable if the authors could calculate and present learning curves for each model employed in this paper. However, I appreciate the fact that training deep learning models could be time consuming and may be not practical to fully implement these calculations. Therefore, this is just a suggestion and it is up to the authors to decide whether to perform such analysis or not.

Authors' response

We thank the reviewer for this constructive suggestion. We agree that broader split-ratio experiments and repeated resampling analyses would be valuable. However, within the scope of the present revision, we prioritised two additions that directly strengthen reproducibility without requiring a full new experimental matrix across all model families.

First, we added a multi-seed robustness analysis in the main manuscript, which reports performance variability across seeds for key classification and detection models. Second, we added representative learning curves in the Supplementary Materials for the main seed-42 runs under both imbalanced and balanced settings. For classification models, these curves include training loss, validation loss, and validation macro-F1. For detection models, training classification loss is reported; intermediate validation detection curves were not dense enough in the current Ultralytics configuration to support a meaningful epoch-by-epoch interpretation.

We did not add a full set of alternative train/test split-ratio experiments in this revision. To address this limitation transparently, we also moderated the claims of generalisability and made the need for broader multi-site and cross-split validation explicit in the revised Discussion.

Changes made in the manuscript

- Results expanded with the multi-seed subsection “Robustness across random seeds” and Table 9.
- Supplementary Materials expanded with representative learning curves for balanced and imbalanced runs.
- Discussion limitations expanded to state explicitly that broader cross-split and out-of-domain validation remain future work.

Comment 3

Reviewer comment

I also highly recommend the authors to share their code in public domain. It is the only way to ensure that others can repeat the same methodology as the authors had intended. Another reason is that the networks employed by this paper are all available in public domain and therefore it is fair to share with the research based on these networks with the public.

Authors’ response

We thank the reviewer for this important recommendation. We agree that public code availability is essential for reproducibility.

Accordingly, the revised manuscript now includes a dedicated “Code availability” section stating that the code used for data preprocessing, model training, evaluation, and figure generation is publicly available in a GitHub repository.

Changes made in the manuscript

- A dedicated “Code availability” section was added at the end of the manuscript.

Minor issue 1

Reviewer comment

In Introduction, 2nd paragraph " ... due to relative modest but highly complex datasets...", modest what? sample size?

Authors’ response

We thank the reviewer. The wording has been corrected for clarity. The revised Introduction now uses the expression “modest sample sizes” to avoid ambiguity.

Changes made in the manuscript

- Introduction revised in the paragraph beginning “Despite recent progress in computer vision applications to archaeology and cultural heritage...”.

Minor issue 2

Reviewer comment

References to DETR and RT-DETR should be used earlier when they were firstly introduced in the main text.

Authors' response

We thank the reviewer. We revised the Introduction so that DETR and RT-DETR are now introduced at first occurrence, with their methodological role explained before their later use in the manuscript.

Changes made in the manuscript

- Introduction revised in the paragraph beginning "DETR-type detectors approach object detection...".

Conclusion

We thank the reviewers once again for their constructive and detailed feedback. The remarks have made it possible to orient the manuscript with more precision, clarity, methodological rigor and disciplinary positioning. Beyond the improvements on the technical part, we have worked on the contribution of protocols for ceramology. The work of the ceramologist, in these different stages, has been considered and explained more clearly in this manuscript so that it can also be used for other specialties of archaeology (studies of ecofacts, small objects, human or animal bones, etc.). We hope that these revisions will satisfactorily address the concerns of the reviewers and that the manuscript will now be suitable for publication.